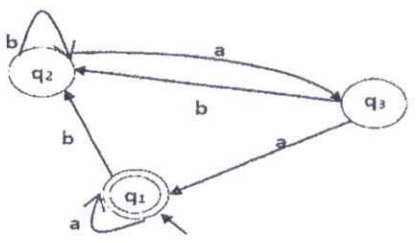
Roll Number: _____ Name_____ Group_____

THAPAR INSTITUTE OF ENGINEERING &TECHNOLOGY, PATIALA
Department of Computer Science & Engineering
Theory of Computation (UCS701) – Auxiliary Examination
Date: 17/02/2026 & Time: 05:30PM MM: 100 & MT: 180 Min

Faculty Name: Dr. Chinmaya Panigrahy

Attempt any four of the five questions. Attempt/Answer all sub-parts like (a), (b), (c) for each question in one place. Do mention Page No. of your attempt on the front page. Question paper is of two pages. Please turn over this page to see the questions on the next page.

Ques. No		Questions	Mark
Q1.	(a)	Apply Thompson's construction to construct an NFA with epsilon moves for the regular expression, (aa+bab)* (b+abb)*.	7
	(b)	Design a DFA that accepts all the binary strings containing 000 as a substring.	7
	(c)	What is the difference between a deterministic finite automaton and a non-deterministic finite automaton. Describe with an example.	6
	(d)	List all the minimal length strings not in the language represented by the regular expression, a*(ab)*b*.	5
Q2.	(a)	Describe various variants of Turing machine in detail.	12
	(b)	Prove that the regular languages are closed under intersection operation.	5
	(c)	Design a Turing machine to add two integers.	8
Q3.	(a)	Design a Post machine for the following language. $L= \{a^n b^n: n>=1\}$.	8
	(b)	Construct a Mealy machine to print residue-modulo 3. Convert the above Mealy machine to an equivalent Moore machine.	10
	(c)	Find the regular expression corresponding to the given finite automata using Arden's theorem. 	7
Q4.	(a)	Prove that the language $L= \{a^n b^n: n>=1\}$ is not regular using Pumping lemma for regular languages.	6
	(b)	Design a Pushdown automata for the following language. $L= \{a^n b^n: n>=1\}$ by constructing (i) Flow diagram (ii) Transition diagram (iii) Tuple representation with transitions	15
	(c)	Write a regular expression to represent all the strings over {0,1,2} that contain at most two 0's.	4

Q5.	**(a)**	Describe the steps to simplify a context-free grammar.	**8**
	(b)	Draw a schematic diagram to describe the Chomsky classification of grammars. Give an example of each type of grammar.	**5**
	(c)	Convert the following CFG into Chomsky Normal Form. $S \rightarrow aXbX$ $X \rightarrow aY \mid bY \mid \varepsilon$ $Y \rightarrow X \mid c$	**12**

NOTE: Answer sheets will be shown on February 18, 2026 at 04:30 PM in L521, CSED Bulding.